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Computing Multiscale Hamiltonian Models from Data

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Abstract

We propose a framework to model, estimate from limited data, and sample high-dimensional non-gaussian multiscale maximum entropy distributions. Its main features are:

1. **Access to an estimated analytical effective hamiltonian model in high-dimension: An interpretable and structured linearly parametrised model of the effective hamiltonian, with coupling parameters, calculation of the normalising constant and access to the entropy value.**
2. **Suitability on reduced datasets: maximum entropy distributions that avoid memorisation**
3. **Sparse modelling of rare events**

The method works as follow:

- **Estimation and sampling. A transport algorithm with guarantees that works over limited datasets, combined with a smoothing procedure across the transport to accurately estimate the normalised energy and entropy. Our method increases by 2 orders of magnitude the accuracy of hamiltonian estimators, giving access to the same quality of estimation for datasets 2 orders of magnitude smaller than what could be done before.**
- **Multiscale modelling. Low-dimensional wavelet spectra moments built from a complex wavelet transform, capturing interactions between scales**
- **Sparse modelling of rare events. A sparse learned modelling of physically relevant rare events**

Application. 1d trajectories of lagrangian particles in 3d turbulence with rare events

Keywords

Abbreviations

Significance statement

Draft

Introduction

Interpretable data-driven probabilistic models for physics

- Building models of high-dimensional physical fields $X \in \mathbb{R}^d$ from training data $(x_i)_{i=1}^n$ seemed impossible up to recently, especially for multiscale fields including long range interactions. Indeed, the calculations of such models face a "curse of dimensionality". Estimating a model by maximum likelihood requires sampling it at each step, usually with Markov Chain Monte Carlo (MCMC) methods, which become prohibitively slow in high dimension because of energy barriers.
- Yet, spectacular results have been obtained by generative AI algorithms. State of the art algorithms are based on score diffusion and flow matching with deep neural networks. These models provide prediction and simulation tools which have considerable applications. However, deep neural network models rely on millions or billions of parameters which are not interpretable, and simulations alone do not improve our understanding of physical phenomena. An outstanding issue is to improve our understanding of physics from models estimated from data.
- Moreover, since these models include a considerable amount of parameters, they require a very large amount of data to generalize. If not enough data is available then they just memorize the training data.
- The generalisation issue is particularly acute for rare events which require much more data to be reproduced. Yet rare events play a crucial role in the physics and applications, for example in meteorology.

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- For systems at equilibrium, we introduce statistical physics models, which can be learned from data, which addresses these issues. Such models provide a bridge between state of the art flow matching algorithms and the calculation of Hamiltonian with interpretable multiscale coupling parameters over scalar potentials and high order moments computed with wavelet transforms. For non equilibrum systems it defines effective Hamiltonian.
- We introduce a representation of rare events with interaction potentials defining sparse representations of physical fields.

Statistical Physics Models with Hamiltonians

- Statistical physics of systems at equilibrium have a Gibbs distribution whose density is defined by their Hamiltonian $p(x) = Z_\theta^{-1} e^{-H(x)}$. For non-equilibrium systems, at each time t the effective Hamiltonian H_t defines a stystem at equilibrium having the same probability distribution.

  Most interpretable models are linear. The Hamiltonian models H_θ is represented over a finite number r of interaction potentials $\phi = (\phi_k)_{k\leq r}$ which have physical interpretations, with a vector of coupling parameters $\theta = (\theta_k)_{k\leq r}$

$$p_\theta = Z_\theta^{-1} e^{-H_\theta} \quad \text{with} \quad H_\theta(x) = \theta^T \phi = \sum_{k\leq r} \theta_k \phi_k(x) \ .$$

  It is an exponential distribution.
- Building interpretable models in a statistical framework means be able to define structured potentials $\phi(x) \in \mathbb{R}^r$ and estimate these coupling vector $\theta \in \mathbb{R}^r$, allowing to analyse physical properties.
- Such models can also be interpreted as maximum entropy distributions conditioned by r observables $\mathbb{E}(\phi)$ which can be estimated from training data. Statistical physics models do not suffer from memorisation because the maximisation of entropy insures that the model has a maximum diversity. It does not memorise the training data.
- Building such Hamiltonian models involves three difficult problems which are affected by the curse of dimensionality:

  1. Sampling p_θ for high-dimensional non-convex Hamiltonian H_θ. Critical issue of energy barriers, particularly acute in high dimension. Standard approach based on MCMC.
  2. Linear Hamiltonian model with a vector ϕ of structured interaction potentials of dimension r which have a physical meaning and providing an accurate model. It relies on priori information on the physics but can also be learned from data. Standard approach based on scalar potentials and high order moments. In high dimension r grows polynomially in d. An outstanding issue is to build lower dimensional models.
  3. Accurate estimation of the Hamiltonian model with the coupling constant θ from a limited set of training samples. Calculated by maximum likelihood, whose gradient requires sampling p_θ at each step, inheriting Problem 1's cost.

  – Classical inverse statistical physics estimates the couplings of low-order Hamiltonians by maximum likelihood, from Boltzmann machines (**?**) and inverse Ising and Potts models (**?**) to FRAME (Filters, Random Fields and Maximum Entropy) texture models (**?**). The likelihood gradient $\mathbb{E}_p[\phi] - \mathbb{E}_{p_\theta}[\phi]$ requires sampling p_θ at each iteration, so Problem 3 inherits Problem 1: for non-convex H_θ, Markov chains remain trapped in metastable states and bias the estimate. Mean-field and TAP approximations (**?**), pseudo-likelihood (**? ?**) and cluster expansions (**?**) remove this inner loop, at the price of approximations valid at weak coupling or of a loss of statistical efficiency. These models are mostly restricted to pairwise potentials, $r = O(d^2)$, whereas the third- and fourth-order interactions carrying scale dependencies would require $O(d^3)$ and $O(d^4)$ couplings. Even when θ is known, the partition function $\log Z_\theta$, and hence the entropy, is computed by thermodynamic integration (**? ?**) or annealed importance sampling (**?**), which interpolate between a tractable reference and p_θ through a sequence of MCMC simulations and face the same energy barriers.
  – Score matching (**?**) eliminates Z_θ by fitting $\nabla_x \log p_\theta$, with a closed-form estimator for linear models. The difficulty is converted rather than removed: its efficiency relative to maximum likelihood is controlled by isoperimetric constants of p_θ (**?**), which also govern the mixing of Langevin dynamics and grow exponentially with energy barriers.
  – Diffusion and flow matching models (**? ? ? ? ?**) learn the scores of noise-perturbed distributions, which are unimodal at high noise, so the transport bypasses energy barriers and addresses Problem 1 in high dimension. Likelihoods and even normalised energies can be computed (**? ?**), but the model is a deep network without a structured Hamiltonian, trained on large datasets. Conversely, maximum entropy models over wavelet scattering moments address Problem 2, reducing r to $O(\log^3 d)$ (**? ?**), but they are sampled by microcanonical gradient descent and provide no coupling parameters. Moment Guided Diffusion (**?**) combines both, and estimates couplings θ_t along the transport; being score-matching estimators, they inherit the variance above.

Main Contributions We compute, from limited data, an explicit Hamiltonian $H_{\theta^*} = \theta^{*\top}\phi$ of high-dimensional multiscale fields, together with its coupling parameters, partition function and entropy. Its coupling parameters quantify interactions across scales as well as the rare extreme events.

– **Estimation of the Hamiltonian.** Coupling parameters estimated along the Moment Guided Diffusion transport have a large variance for non-convex energies. We reduce it with a regularisation across the transport, adapted from dual score matching (**?**), which improves the accuracy of the estimated energy by two orders of magnitude on controlled examples. The partition function $\log Z_{\theta^*}$ and the entropy are then computed by a thermodynamic integration along the transport, which requires no MCMC.

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– **Potentials for rare events.** Rare events are too scarce in training data to be learned without strong priors. We represent them with a dictionary of waveforms in which they are sparse, and model the marginals of the resulting coefficients with windowed exponential potentials. This adds $O(\log d)$ coupling parameters to the scattering spectra potentials (**?**), which encode high-order multiscale interactions with $O(\log^3 d)$ moments.

We apply this methodology to one-dimensional velocity trajectories of Lagrangian particles in three-dimensional turbulence, where rare events occur at all scales.

Method

Sampling in high dimension

AI generative models based on score diffusions and flow matching are able to sample high-dimensional complex distributions having a non-convex energy with a homotopic approach implemented with an SDE, which implements a transports from an isotropic Gaussian distribution towards the data distribution. The high-dimensional velocity is computed with a deep neural network which have typically at least millions of parameters. Lower dimensional models have recently been introduced with Moment Guided Diffusion, with a velocity flow which has an analytical expression computed from a limited number of energy potentials. It thus avoids using a deep neural network to estimate the model, allowing to considerably reduce the number of training examples. Next section briefly reviews this MGD algorithm.

Although score diffusion algorithms can sample probability distributions, they are unable to provide accurate estimations of the energy parameters. We shall explain that this is due to the non-convexity of the Gibbs energies, which have a log-Sobolev constants which grow exponentially with the energy barriers (1). In the following section, we introduce a procedure eliminate this huge variance term through a regularisation across the transport, used to estimate energies calculated by deep neural networks. It provides, for the first time an estimation of high-dimensional coupling parameters of physical Gibbs energies.

Sampling by Flow Matching

– Flow matching algorithms learn a sampler for data by learning a homotopic transport between noise and data. This path successfully avoids the potential barriers that plague MCMC algorithms, by placing particles within modes at early times when barriers are down.
– A variance-preserving stochastic interpolant I_t:

$$I_t = \cos(\alpha_t)\, Z + \sin(\alpha_t)\, X, \quad t \in [0,1],$$

– The key observation is that the PDF $q_t(x)$ of the interpolant I_t can be sampled via an SDE whose coefficients are estimable from data, with a bias term b_t which adjusts the probability distribution of X_t to q_t and a second term $\nabla \log q_t$ which compensate the Brownian noise introduced in the SDE:

$$dX_t = b_t(X_t)\, dt + \sigma^2 \nabla \log q_t(X_t)\, dt + \sqrt{2}\,\sigma\, dW_t,$$

where $b_t(x) = \mathbb{E}\big[\frac{d}{dt} I_t | I_t = x\big]$.
– Since a conditional expectation is the minimizer of a quadratic loss, b_t and $\nabla \log q_t$ can be learned by minimising this loss. This is an infinite-dimensional estimation problem for the energy $-\log q_t$. It came as a surprise that such high-dimensional velocities could be estimated with a deep neural network but it requires large deep networks which involve at least millions of parameters. Large training data sets are needed to avoid a simple memorisation of the training data set.
– Modelling $-\log q_t$ with a low-dimensional exponential model $-\log p_\theta$ can lead to a huge uncontrolled error. In this case, the transport does not sample from p_θ. Score diffusion only works if $\log q_t$ is well approximated.

Sampling with Moment Guided Diffusion

– MGD is a flow matching that does not aim at reproducing the distribution p_t of I_t, but only the finite set of moments

$$m_t = \mathbb{E}(\phi(I_t)).$$

The underlying models of q_t have a Gibbs energy of the form $\theta_t^T \phi$. In this case, the score term $\nabla \log q_t$ in the SDE (2) is replaced by an analytical expression $\theta_t^T \nabla \phi$. In MGD, the replacement for the bias term b_t adjusts the moments instead of the whole probability distribution and can similarly be written $\eta_t^T \nabla \phi$. It leads to a closed-form, finite-dimensional transport:

$$dX_t = \big(\eta_t^\top - \sigma^2 \theta_t^\top\big) \nabla \phi(X_t)\, dt + \sqrt{2}\,\sigma\, dW_t, \quad X_0 = Z,$$

where W_t is a Brownian noise. The constant η_t adjust the velocity to match the modification of the moments m_t

$$G_t\, \eta_t = \frac{d}{dt} m_t,$$

whereas θ_t is a score parameter which eliminates the Brownian noise of the SDE with a score matching formula

$$G_t\, \theta_t = \mathbb{E}\big[\Delta \phi(X_t)\big],$$

where G_t is the Gram matrix

$$G_t = \mathbb{E}\big[\nabla \phi(X_t) \cdot \nabla \phi(X_t)^\top\big]. \tag{1}$$

If this coupled system admits a solution, then the moment condition $\mathbb{E}\big[\phi(X_t)\big] = \mathbb{E}\big[\phi(I_t)\big]$ holds for all $t \in [0,1]$.
– This is a non-linear McKean–Vlasov equation, discretised with an iterative particle algorithm designed to enforce empirical moment constraints (thus avoiding numerical instabilities that lead to divergence of particles).
– The main difference with flow matching is that estimation of parameters is done inline, while sampling, iteratively from $t = 0$ to $t = 1$, while flow matching estimation can be done offline and estimations of $\log p_t$ at different time can be treated in parallel.
– The parameter θ_t is an score matching estimation computed over the distribution p_t of X_t, with moments m_t. It defines an underlying maximum entropy distribution $p_{\theta_t} Z_{\theta_t}^{-1} e^{\theta_t^T \phi}$ that converges towards the maximum entropy distribution conditioned by the moments m_t when the volatility σ goes to infinity.

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- Estimators of θ_t have a very large variance which grows exponentially with the amplitude of energy barriers. This variance comes from the fact that this estimation results from a minimisation of a Fisher divergence which depends on $\nabla \log p_t$. The gradient eliminates the influence of the normalisation constant Z_{θ_t} but it introduces instabilities in the estimation, as opposed to estimators which minimize a Kullback-Leibler divergence based on $\log p_t$, but which are much more costly to compute because they depend on Z_{θ_t}.
- This limitation is intrinsic to the criterion rather than to our implementation. Score matching is provably less statistically efficient than maximum likelihood by a factor controlled by the same functional constant that governs the mixing of Langevin dynamics (1). For a multimodal density whose modes are separated by barriers of height ΔE in the energy $E = -\log p$, this constant grows like $e^{\Delta E}$. Sampling difficulty and estimation variance are therefore two consequences of a single property of the energy landscape.
- Replacing G_t by an empirical covariance matrix gives very high variance estimations of coupling parameters because the inversion of G_t has instabilities. It yields a poor estimation of the energy $\theta_t^T \phi$ in particular when $t = 1$ which is the final energy we want to estimate.
- In the following we shall assume that $\mathbb{E}(\phi\phi^T)$ is a well conditioned matrix so that the accuracy of the estimation of θ_t^T is equivalent to the accuracy of the estimation of $\theta_t^T \phi$.

Hamiltonian and coupling parameters estimation

- Obtaining accurate empirical estimations of hamiltonians from limited datasets is a outstanding challenge for multimodal distributions, especially the high dimensional distributions encountered in physics.
- It means obtaining an estimator $\hat{\theta}$ of parameters θ_* over n realisations such that the Kullback Leibler divergence $\mathbb{E}\big[D_{\mathrm{KL}}(p_{\theta_*} \,\|\, p_{\hat{\theta}})\big]$ is small for small values of n.
- Due to central limit theorem, the following behaviour is expected $\mathbb{E}\big[D_{\mathrm{KL}}(p_{\theta_*} \,\|\, p_{\hat{\theta}})\big] \sim \frac{C}{n}$ when n goes to infinity. Small $\mathbb{E}\big[D_{\mathrm{KL}}(p_{\theta_*} \,\|\, p_{\hat{\theta}})\big]$ when n is small means small n.
- This may be achieved for MGD by averaging estimators over the homotopic path.

Accurate estimate of the hamiltonian

- Numerically accessible estimators of energy are computed with score matching are not accurate, they have high variance, which is exponential in the depth of the wells, so $\mathbb{E}\big[D_{\mathrm{KL}}(p_{\theta_*} \,\|\, p_{\hat{\theta}})\big]$ is large. It manifest by the incapacity of theses estimators to capture relative amplitudes between modes.
- Yet, in both flow matching and in MGD in our case, integrations of estimated Fisher divergences provides upper bound over Kullback-Lieber divergence between the estimated distribution, from the transport, and target distribution. Using Girsanov's formula, we may write

$$\mathbb{E}\big[D_{\mathrm{KL}}(p_t \,\|\, \hat{p}_t)\big] \le \int_0^1 \mathbb{E}\big[\|\hat{\theta}_t^\top \nabla\phi(X_t) + \nabla \log p_t(X_t)\|^2\big]dt + \tfrac{1}{4\sigma^2}\int_0^1 \mathbb{E}\big[\|\hat{\eta}_t^\top \nabla\phi(X_t) - \eta_t^\top \nabla\phi(X_t)\|^2\big]dt \tag{2}$$

where $\hat{p}_t$ is the estimated distribution from MGD with estimated parameters $\hat{\theta}_t$ and $\hat{\eta}_t$. Expectation are taken with respect to X_t and different realisations of the estimators.

- This suggest that high variance estimators $\hat{\theta}_t^\top \phi$ can be regularised by integrating all score matching estimation in time.
- This strategy was developed with the dual score-matching objective in Guth et al. The time regularity of the estimates of the hamiltonian is imposed by imposing a constraint over its derivate over time. It amounts to not only estimate the score $\nabla \log p_t$, but also an estimate of the time derivative $\frac{d}{dt}\log p_t$. Learning $\frac{d}{dt}\log p_t$ couples the energy across diffusion time, propagating the relative amplitude of the modes from $t = 0$, where the distribution is unimodal, to $t = 1$, where single-time score-matching estimators see each mode locally and cannot recover their relative weights.
- For MGD, this strategy is implemented by a time-regularised hamiltonian $\tilde{\theta}_t^\top \phi$ in place of $\theta_t^\top \phi$. Since MGD is inline—the particles X_t are produced while solving the SDE—this is done a posteriori from a first solve, fitting $\tilde{\theta}_t$ from the particles X_t.
- The dual score-matching objective combines the traditional score-matching loss, optimised both in flow matching and in MGD, with a time regularised term, which imposes parameter sharing. For MGD and maximum entropy models, the two terms are integrates over time

$$\mathcal{L}_{\mathrm{DSM}}(\tilde{\theta}) = \int_0^1 \mathbb{E}\Big[\,\big\|\tilde{\theta}_t^\top \nabla\phi(X_t) + \nabla \log p_t(X_t)\big\|^2\,\Big]\,dt. \tag{3}$$

$$\mathcal{L}_{\mathrm{TSM}}(\tilde{\theta}) = \int_0^1 \mathbb{E}\left[\,\left\|\tfrac{d}{dt}\tilde{\theta}_t^\top\big(\phi(X_t) - m_t\big) + \tfrac{d}{dt}\log p_t(X_t)\right\|^2\,\right]dt. \tag{4}$$

where p_t is the distribution of X_t obtained from MGD.

- Adding $\mathcal{L}_{\mathrm{TSM}}$ to $\mathcal{L}_{\mathrm{DSM}}$ enforces an energy $\tilde{\theta}_t^\top \phi$ that varies smoothly in time, reducing the local-in-time statistical fluctuations of the pointwise estimate $\hat{\theta}_t^\top \phi$.
- Although $\log p_t$ is not tractable, we prove in Appendix 2 that both losses can be calculated analytically up to an additive constant:

$$\mathcal{L}_{\mathrm{DSM}}(\tilde{\theta}) = \int_0^1 \Big(\tilde{\theta}_t^\top\, G_t \tilde{\theta}_t \;-\; 2\,\tilde{\theta}_t^\top\, \mathbb{E}[\Delta\phi(X_t)]\Big)\,dt \;+\; \mathrm{cst}, \tag{5}$$

$$\mathcal{L}_{\mathrm{TSM}}(\tilde{\theta}) = \int_0^1 \Big(\tfrac{d}{dt}\tilde{\theta}_t^\top\, C_t\, \tfrac{d}{dt}\tilde{\theta}_t \;+\; 2\,\tfrac{d}{dt}\tilde{\theta}_t^\top\, \tfrac{d}{dt} m_t\Big)\,dt \;+\; \mathrm{cst} \tag{6}$$

with covariance matrix $C_t = \mathbb{E}\Big[\big(\phi(X_t) - m_t\big)\big(\phi(X_t) - m_t\big)^\top\Big]$.

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– The regularised parameters are then defined as

$$\tilde{\theta} = \arg\min_{\tilde{\theta}} \ \mathcal{L}_{\mathrm{DSM}}(\tilde{\theta}) + \lambda \mathcal{L}_{\mathrm{TSM}}(\tilde{\theta})$$

for a positive weight λ. This loss does not vanish unless X_t is of maximum entropy, which only occurs as $\sigma \to \infty$; otherwise $\tilde{\theta} \neq \theta$. For finite σ, $\hat{\tilde{\theta}}_1$ is a biased estimator of θ_*. Nevertheless, as $\sigma \to \infty$, $\tilde{\theta}_t$ converges towards the Lagrange multipliers of the maximum entropy distribution with moments m_t.

– We expect the estimated energies $\hat{\tilde{\theta}}_t^\top \phi$ to have lower variance than the MGD estimates $\hat{\theta}_t^\top \phi$.

Numerical estimation of energies for multimodal distributions

– As said, obtaining accurate empirical estimations of energies from limited datasets is extremely hard for multimodal distributions.

– At fixed dataset size, the previous section predicts an error which grows like $e^{\Delta E}$ in the barrier height. We verify this numerically, and measure by how much the time regularisation reduces it.

– We measure the quality of an estimator $\hat{\theta}$ by computing the expectation $\mathbb{E}(D_{\mathrm{KL}}(p_{\theta_*}|p_{\hat{\theta}}))$, which is the average error in Kullback Leibler divergence.

– Considering a dataset of fixed size, the error in KL estimation increases exponentially with the depths of the multimodal distribution wells.

– We consider a generic synthetic unbalanced bimodal maximum entropy distribution

$$p_\theta(x) = Z_\theta^{-1} e^{-\theta^\top \phi(x)}$$

where $\phi(x) = (x, x^2, x^3, x^4)$ and $\theta = |\theta|(1/2, -5, 0, 1)$. The depth of the well is proportional to $|\theta|$.

– Over different runs of MGD, over different realisations $(x_i)_{i=1}^n$, we can estimate the maximum entropy model $\theta_1^\top \phi(x)$ and the regularised $\tilde{\theta}_1^\top \phi(x)$, and compare the Kullback Leibler error $\mathbb{E}\big[D_{\mathrm{KL}}(p_{\theta_*} \| p_{\hat{\theta}_n})\big]$, for different values of θ. This is shown Figure 1.

– The KL error computed with MGD grows exponentially in $|\theta|$, which matches the theoretical prediction. Our regularisation procedure leads to an error that grows with $|\theta|$ too, yet the error is two order of magnitude lower than without regularisation. The same quality of estimator can be obtained from MGD with n realisations, or with $100^{-2}n$ realisations and the regularisation.

– The quality has to be compared to the Cramér–Rao bound $\mathrm{Cov}(\hat{\theta}) \geq n^{-1}\mathrm{Cov}_{p_\theta}(\phi)^{-1}$, which sets the minimal variance of an energy estimation.

If we have an unbiased estimator, then a calculation in Appendix 2 of a Taylor expansion, when n goes to infinity, of $\mathbb{E}(D_{\mathrm{KL}}(p_{\theta_*}|p_{\hat{\theta}}))$, combined with the Cramér Rao bounds gives

$$\mathbb{E}\big[D_{\mathrm{KL}}(p_\theta \,\|\, p_{\hat{\theta}})\big] \ \geq \ \frac{r}{2n} + o(n^{-1}). \tag{7}$$

where r is the number of parameters. For all estimators, $\mathbb{E}\big[D_{\mathrm{KL}}(p_\theta \,\|\, p_{\hat{\theta}})\big]$ the expected KL should decrease like $1/n$ in the limit of large n due to the central limit theorem.

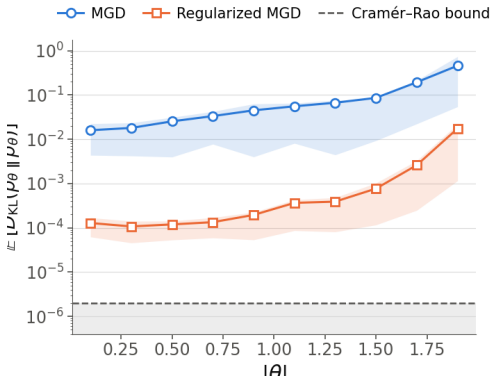


Figure 1 Accuracy of the energy estimator, measured by the mean Kullback–Leibler divergence $\mathbb{E}\big[D_{\mathrm{KL}}(p_{\theta_*}, p_{\hat{\theta}})\big]$, as a function of $|\theta|$ for the target density $p(x) = Z_\theta^{-1}\, e^{-|\theta|(x^4 - 5x^2 - x/2)}$ with $n = 10^6$ samples. Blue: MGD estimator. Orange: regularised estimator. Dotted Black: Theoretical Cramer-Rao lower bound on $\mathbb{E}\big[D_{\mathrm{KL}}(p_{\theta_*}, p_{\hat{\theta}})\big]$, equal to $r/2n$, with $r = 4$ here.

– The regularisation does not lead to an optimal estimator, as the Cramer Rao bound is constant with $|\theta|$ and 2 orders of magnitudes smaller that the error with regularised estimators at fixed n.

Partition function and Entropy Calculation

– A physical model of the hamiltonian also requires its thermodynamic quantity. The partition function sets the absolute scale of the energy $U_{\theta^*} = -\log p_{\theta^*} = H_{\theta^*} + \log Z_{\theta^*}$, and allows models to be compared. It is a high-dimensional integral that cannot be estimated directly.

– MGD reduces it a to one-dimensional integral along the transport.

– We recall that $\frac{d}{dt}\log Z_{\theta_1} = \frac{d}{dt}\theta_1^\top \mathbb{E}_{p_{\theta_t}}(\phi)$, which may be integrated from 0 to 1 to compute $\log Z_{\theta_1}$. Unfortunately, $\mathbb{E}_{p_{\theta_t}}(\phi)$ is not tractable. Replacing it with m_t transform the inequality into a lower bound over the normalising constant

$$\begin{aligned}\log Z_{\theta_1} \ \geq \ \log Z_\sigma \ &\underset{\text{def}}{:=} \ \log Z_{\theta_0} + \theta_0^\top m_0 - \theta_1^\top m_1 \\ &\quad + \int_0^1 \theta_t^\top \tfrac{d}{dt} m_t \, dt,\end{aligned} \tag{8}$$

Similarly, than in MGD, we conjecture that, as $\sigma \to \infty$,

$$\log Z_\sigma \to \log Z_{\theta^*}. \tag{9}$$

Conjecture: we expect quadratic convergence: they exists C such that

$$|\log Z_\sigma - \log Z_{\theta^*}| \leq C\sigma^{-2}.$$

– We may also compute an entropy lower bound, similar to the one in MGD for $S(p_t)$, using that $S(p_{\theta_t}) = \log Z_{\theta_t} + \theta_t^\top m_t$

$$S(p_{\theta_1}) \geq S_\sigma = S(p_{\theta_0}) + \int_0^1 \theta_t^\top \tfrac{d}{dt} m_t \, dt. \tag{10}$$

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Remark 1 *this is a lower bound over the maximum entropy distribution $S(p_{\theta_1})$ while the lower bound in MGD's paper is on $S(p_{\theta_1})$ where p_1 is the distribution of X_1.*

– The entropy lower bound can be directly estimated from MGD, because the variance in estimators is averaged over time, but $\log Z_\sigma$ cannot: it depends on the energy $\theta_1^\top \phi$, whose high variance makes the raw MGD estimate of $\log Z_\sigma$ too imprecise. We compute it instead from the regularised energy $\tilde{\theta}_t^\top \phi$ of Section **??**.

Low-Dimensional Multiscale Interaction Potentials

– A central challenge, addressed in this section, is to construct a low-dimensional and interpretable energy vector $\phi = (\phi_k)_{k \le r}$ providing an accurate linear expansion of the Hamiltonian. We rely on prior physical information to reduce the dimension from d to $O(\log^3 d)$.
– Two requirements guide the construction. First, the maximum entropy distribution must exist. The Gibbs form is a distribution only if it is normalisable, so the couplings are restricted to the convex domain

$$\Theta = \Big\{\theta \in \mathbb{R}^r : Z_\theta = \int e^{-\theta^\top \phi(x)}\, dx < \infty\Big\},$$

which is generally not all of $\mathbb{R}^r$. If some $\theta \in \Theta$ satisfies $\mathbb{E}_{p_\theta}(\phi) = \mathbb{E}_p(\phi)$, then p_θ is the maximum entropy distribution; the converse holds when Θ is open and $\mathbb{E}_p(\phi)$ lies in the interior of the set of achievable moments (**?**). A simple sufficient condition is available when a leading potential ϕ_{k_0} dominates the others at infinity: then $\{\theta : \theta_{k_0} > 0\} \subseteq \Theta$, and existence is guaranteed by checking the sign of this single coupling. For instance, ϕ_{k_0} can be a high order monomial such as $\sum_t x(t)^6$, as in our applications. Second, we normalise $\phi(0) = 0$, hence $H_\theta(0) = 0$, as in Landau–Ginzburg φ^4 theories.
– Since $H_{\theta^*}(0) = 0$, the mean energy decomposes as $\mathbb{E}(H_{\theta^*}) = \sum_{k \le r} \theta_k^* \, \mathbb{E}(\phi_k)$, where each term measures the contribution of the potential ϕ_k.
– Interpretability in scalar field theory calls for scalar potentials expressed as polynomial expansions. Moments of order 2 characterise Gaussian structure, while orders 3 and 4 capture the non-Gaussian scale dependencies typical of physical fields.
– For stationary fields, the number of such moments computed in the Fourier domain grows as d^2 for the bispectrum and d^3 for the trispectrum, well beyond the size of realistic training sets. Wavelet scattering moments exploit fast spatial decorrelation of wavelet coefficients to reduce these counts to $O(\log^2 d)$ and $O(\log^3 d)$ respectively. Their construction is reviewed in the next section.
– Rare events are harder to model because their occurrence in the dataset is small, so stronger priors are needed. We capture them through sparse representations, which first requires isolating rare events into a dedicated class. Sparsity is quantified by windowed ℓ^β norms for $\beta \le 1$, from which we derive suitable potentials.
– In summary, we use scalar potentials together with a compressed polynomial expansion up to order four, both defined over a wavelet transform that separates scales and over a dictionary that sparsifies extreme events.

Wavelet High Order Interaction Potentials

A complex wavelet transform performs a separation of the field variations at different scales. We briefly review the properties of a dyadic complex wavelet transform, the distribution of wavelet coefficients, and introduce $3rd$ and $4th$ order scattering moments which capture dependencies across scales with $O(\log d^3)$ coefficients. The choice of these moments can also enforce particular symmetries of the probability distribution.

Complex wavelet transform

In the following we consider one-dimensional wavelets but the extension to multidimensional signals is studied in (**?**).

– A wavelet is a localized waveform (in time and frequency) $\psi(t)$, $t \in \mathbb{R}^{\not=}$, with zero average $\int \psi(t)\, dt = 0$. We use a complex wavelet $\psi(t) = g(t)\, e^{i\xi t}$, where $g(t) = g(-t)$ is a real, regular, symmetric window whose Fourier transform $\hat{g}(k)$ has a support concentrated at low frequencies k. It results that $\hat{\psi}(k) = \hat{g}(k - \xi)$ has a support concentrated around the frequency ξ (e.g. the Morlet wavelet).
– The wavelet is dilated by dyadic scales 2^j and rotated

$$\psi_j(u) = 2^{-j}\, \psi(2^{-j} t),$$

so that $\hat{\psi}_j$ is centered at the frequency $2^{-j}\xi$ with bandwidth $\propto 2^{-j}$.
– Wavelet coefficients are convolutions computed at every time t of the field

$$Wx[j](t) = x * \psi_j(t) \text{ for } j = 0, .., J+1$$

where ψ_{J+1} is a Gaussian dilated at the largest scale 2^J.
– Time–frequency localization: $x * \psi_j(t)$ depends on x in a spatial neighborhood of width $\propto 2^j$ and on $\hat{x}$ in a frequency band of width $\propto 2^{-j}$.

Wavelet Coefficient Marginals

– Modelling the marginal distribution of wavelet coefficients is a well-studied problem, and a natural starting point for building hamiltonians from a wavelet transform. When these coefficients are Gaussian, the marginal is entirely characterised by the second-order moment

$$\sigma_j^2 = \mathbb{E}\big(\text{Ave } |Wx[j]|^2\big),$$

where the operator Ave(.) performs the average over the signal dimension d.
– Many physical and financial processes are however *intermittent*: their activity is not uniform but concentrated in localised, irregular bursts. This translates into non-Gaussian marginals with fat tails and a sharp peak at zero: most realisations are small, punctuated by

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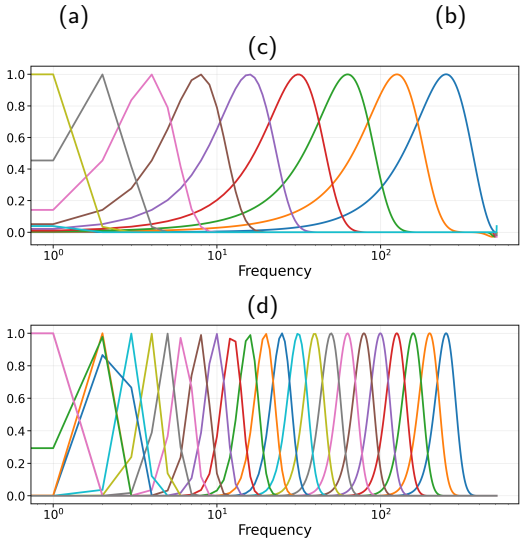


Figure 2 (a) One-dimensional Morlet wavelet $\psi(t)$ and (b) one dimension Q-band wavelet (for $Q = 3$) $\chi(t)$. The wavelets are complex functions whose real and imaginary parts are shown respectively in blue and red. The wavelets Fourier transforms squared modulus $(|\hat{\psi}_j(\omega)|^2)_{1\le j\le J}$ (respectively $(|\hat{\chi}_m(\omega)|^2)_{1\le m\le M}$ pave the frequential domain with few overlap between their support. This is illustrated in (c) (respectively (d)).

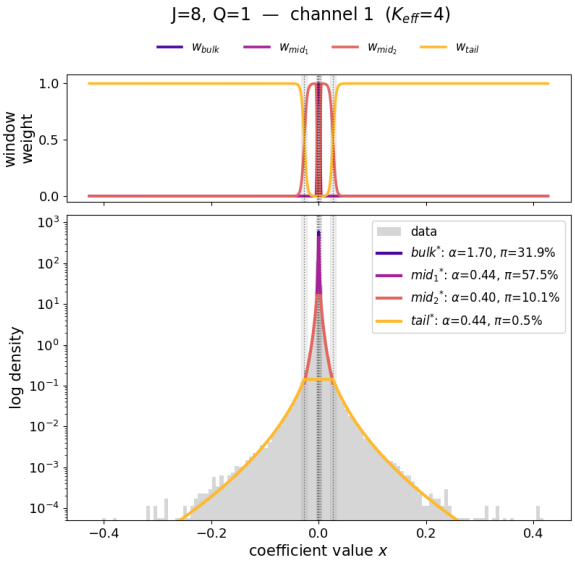


Figure 3 Bulk/tail decomposition of the wavelet-coefficient marginal for a single scale channel j ($Q = 3$), illustrating the construction of Section **??**. **Top:** the soft windows $w_{j,k}(v)$ of (12) partitioning the amplitude axis into K_j regions, $\sum_k w_{j,k}(v) = 1$. Each window is a sigmoid centred at the boundary $c_{j,k}$ with transition width $\delta_{j,k}$ (vertical dotted lines and shaded bands), smoothly gating which region's density dominates at a given $|v|$. **Bottom:** empirical histogram of $|Wx[j]|$ (gray) with the fitted truncated generalised-Gaussian density $p_{j,k}(t) \propto \exp\big(-(|t|/s_{j,k})^{\alpha_{j,k}}\big)$ overlaid per region (coloured curves, matching the window colours above), with shape $\alpha_{j,k}$ and scale $s_{j,k}$ fit by maximum likelihood on the truncated segment $[c_{j,k-1}, c_{j,k})$. The exponents $\alpha_{j,k}$ are those entering (12). $\pi_{j,k}$ denotes the empirical occupancy weight of region k, the fraction of samples falling within its support.

a few large ones. The wavelet coefficients $Wx[j]$ are then sparser than a Gaussian.

- Intermittency is standardly enforced at each scale by prescribing the modulus moment $\mathbb{E}(\text{Ave}\,|Wx[j]|)$, whose ratio to the second-order moment provides a scale-dependent sparsity measure.
- First- and second-order moments alone, however, do not capture strong intermittency: the corresponding maximum entropy distributions have tails decaying no more slowly than Laplacian, and a peak at zero no sharper than Laplacian. This excludes the rare events and fat tails observed in turbulence.
- To capture such fat-tailed marginals, we model the marginal of $|Wx[j]|$ at each scale j as a windowed exponential distribution

$$p_j(v) \;=\; \mathcal{Z}_{\theta_j}^{-1}\,\exp\Big(-\sum_{1\le k\le K_j}\theta_{j,k}\,g_{j,k}(v)\Big), \qquad v\in[0,\infty), \tag{11}$$

whose potential components are windowed power laws of the amplitude,

$$g_{j,k}(v) \;=\; w_{j,k}(v)\,|v|^{\alpha_{j,k}}, \qquad w_{j,k}(v) = w\Big(\frac{|v|-c_{j,k}}{\delta_{j,k}}\Big). \tag{12}$$

Each $g_{j,k}$ is a scalar function of a single coefficient amplitude. The soft window $w_{j,k}$ is a sigmoid centred at the boundary $c_{j,k}$ with transition width $\delta_{j,k}$; it localises the component around that amplitude, and the windows are normalised so that $\sum_{k\le K_j} w_{j,k}(v) = 1$ for all v, so that the K_j components partition the amplitude axis. Small $c_{j,k}$ captures the peak at zero, while large $c_{j,k}$ specifies the tail, and the exponent $\alpha_{j,k}$ controls the local decay rate over the corresponding amplitude range.

- The potential vector of the field is obtained by applying each $g_{j,k}$ pointwise to the wavelet coefficients and averaging over the signal,

$$\phi_{\mathrm{w}}(x) \;=\; \Big(\text{Ave}\,g_{j,k}\big(|Wx[j]|\big)\Big)_{j,k}, \qquad 0\le j\le J{+}1,\;\; 1\le k\le K_j \tag{13}$$

The window parameters $c_{j,k}$, $\delta_{j,k}$ and the exponents $\alpha_{j,k}$ are regressed from data, as described in Appendix **??**, whereas the couplings $\theta_{j,k}$ are estimated by the procedure of Section **??**. Figure 3 illustrates this decomposition for one scale channel.

Interactions between scales with scattering spectra

- The previous modelling treat scales independently. While scale are independent for gaussians process, they are strongly dependent in non gaussian processes.
- Wavelet coefficients $Wx[j] = x * \psi_j$ at different scales have nearly disjoint frequency supports because the Fourier transform of ψ_j and $\psi_{j'}$ have nearly disjoint support. It results that wavelet coefficients are decorrelated across scales $\mathbb{E}\big[Wx[j]\,Wx[j']\big] \approx 0$ for $j \neq j'$. The decorrelation results from the phase which oscillates at different frequencies.
- To reveal these dependencies between scales requires to use high-order moments. This can also be interpreted

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as an elimination of the phase of the phase with $|\bar{x}_j|^2$: it demodulates the oscillation onto a low-frequency envelope whose correlation with coarser scales is non-zero.

– We consider the 3rd and 4th order moments

$$\mathbb{E}(Wx[j](\cdot)\,|Wx[j+k](\cdot-\tau)|^2)$$

$$\mathbb{E}(|Wx[j](\cdot)|^2\,|Wx[j+k](\cdot-\tau)|^2).$$

– Not suitable for numerical applications, because these sets of moments are made of, for stationary processes, $O(d^2)$ and $O(d^3)$, too much to be estimated. need to compress into a lower dimensional set of moments which can be estimated from smaller datasets.
– We explain in appendix that a second wavelet transform can be used to compress these matrices
– A scattering spectra only keeps the diagonal coefficients which are

$$\mathbb{E}(\text{Ave } Wx[j]\,W|Wx|^2[j+k,j])$$

and

$$\mathbb{E}(\text{Ave } W|Wx|^2[j,j+k']\;W|Wx|^2[j+k,j+k']).$$

where $W|Wx|^2[j,j+k'](t) = |x*\psi_j|^2 * \psi_{j+k}(t)$.
– 3rd order moments are approximated from moment function

$$\phi_3(x) = \Big(\text{Ave } Wx[j]\,W|Wx|^2[j+k,j]\Big)_{0\le j\le J,k>0}$$

where k is considered positive to avoid redundancy.
– Similarly, 4th order moments are approximated from

$$\phi_4(x) = \Big(\text{Ave } W|Wx|^2[j,j{+}k']\,W|Wx|^2[j{+}k,j{+}k']\Big)_{0\le j\le J,k>0,k'>k}$$

where k is considered positive to avoid redundancy, and $k' > k$ because taking the modulus or the square of $Wx[j+k]$ shifts it frequential support towards lower frequencies than 2^{-j+k}.
– ϕ_3 and ϕ_4 define a dimensional approximation of 3rd and 4th-order moments, compressed in a wavelet basis: $O(J^3) = O(\log^3 d)$ coefficients.
– Contrarily to previous approaches, we consider scattering covariances taken over the square $|Wx[j]|^2$ and not with respect to the modulus $|Wx[j]|$. This permits to have true 3rd and 4th-order moments with are suitable for physical analysis.
– Symmetry properties do appear from vanishing coupling parameters. If $p(x) = p(-x)$ then the coupling parameters leading the potentials ϕ_3 must be zero. These potentials may in this case be eliminated to define maximum entropy distribution which satisfy the symmetry. If p is invariant to time reversal then one can verify that coupling parameters leading ϕ_3 and ϕ_4 are real. Eliminating the imaginary parts of such moments also define maximum entropy distributions which satisfy this symmetry.

Figure 4 EXPLAIN AND MAKE A FIGURE WHICH ILLUSTRATES ϕ_χ

Rare events with sparse representations

– Include in the dictionary of wavelet marginals a representation that sparsifies extreme events
– We have little data to capture rare events. To estimate them requires strong priors. We define this prior as the ability to define sparse representation of such events. In such representation, rare events concentrate into a few large coefficients, while everything else stays near zero.
– For stationary processes, we use a dictionary of translated waveforms. Waveforms χ_m are chosen such that each convolution

$$\widetilde{W}x[m](t) \;=\; x\star\chi_m(t), \qquad 0\le m\le M, \qquad (14)$$

takes many negligible values and a few large ones when a rare event occurs.
– Equivalently, $x\star\chi_m(t)$ is expected to have a sparse marginal distribution, sharply peaked at zero and with fat tails.
– As for the Morlet coefficients, this cannot be captured by maximum entropy distributions conditioned on first and second order moments, because the peak and the tails decay more slowly than $e^{-|\widetilde{W}x[m](t)|}$. We therefore model the marginal of $|\widetilde{W}x[m]|$ with the same windowed exponential family as in (11)–(12), replacing the scale index j by the dictionary index m,

$$p_m(v) \;=\; \mathcal{Z}_{\theta_m}^{-1}\exp\Big(-\sum_{1\le k\le K_m}\theta_{m,k}\,g_{m,k}(v)\Big), \qquad g_{m,k}(v) = w_{m,k}(v$$

(15)

with the same soft windows $w_{m,k}(v) = w\big((|v| - c_{m,k})/\delta_{m,k}\big)$ summing to one along the amplitude axis. Small $c_{m,k}$ captures the sharp peak, large $c_{m,k}$ specifies the tail, and $\alpha_{m,k}$ its local decay rate. The corresponding potential is

$$\phi_\chi(x) \;=\; \Big(\text{Ave}\; g_{m,k}\big(|\widetilde{W}x[m]|\big)\Big)_{m,k}, \qquad 0\le m\le M,\;\; 1\le k\le K_$$

(16)

with $c_{m,k}$, $\delta_{m,k}$ and $\alpha_{m,k}$ regressed from data as in Appendix **??**. The full scalar potential is the concatenation $(\phi_{\rm w},\phi_\chi)$, so that ψ_j describes the background and χ_m the rare events.

Numerical Results

- Computing θ_* and normalised hamiltonians with enough accuracy over high dimensional non gaussian multiscale processes for applications is an outstanding challenge.
- the originality of our approaches is we can compute the coupling parameters from moments. In our case from a representation of moments of limited dimension that is structured.
- We show that our algorithm scales to high dimension, and can estimate accurately normalised effective hamiltonians of non gaussian multiscale processes observed from limited datasets. Combined with a scattering spectra structured

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modelling, we are able to estimate coupling parameters between scales, opening avenues for physics.
- In addition, having a numerical access to high dimensional integrals defined over these energies like entropy or normalising constant are generic challenges in physics.
- Illustrate over 1d trajectories of 3D lagrangian turbulent particles undergoing extreme accelerations, the same test-bed explored in (?).
- In this particular context, the acces to the coupling parameters opens promising avenues to study the emergence of rare events in turbulence (which is of main interest for weather science). We give preliminary results and show that rare events are not due to interactions across scales, and that interactions between scales are long range.

Modelling rare events in lagrangian turbulence

- In Lagrangian turbulence, particles undergo rare yet extreme accelerations, which appear in the velocity trajectories as large oscillatory events localised in time and scale. They may be separated from the background using a sparse dictionary defined with an adapted wavelet. Turbulence is inherently multiscale, and so are its rare events, which occur at all frequencies, from the inertial range down to the dissipative scales.
- We consider the dataset from (?), obtained by tracking passive tracer particles through a direct numerical simulation of three-dimensional, statistically stationary, homogeneous and isotropic incompressible turbulence, computed on a spatial grid of 1024^3 points in a triply periodic cube. As each particle is advected by the flow, its velocity is recorded along its path, so that every Lagrangian trajectory yields one time series per velocity component. These series are sampled finely enough in time to resolve the Kolmogorov time τ_η, the dissipative scale below which viscosity damps velocity fluctuations and terminates the energy cascade. With $\tau_\eta \approx 10$ time steps, the trajectories show a clear separation between the integral time and τ_η, spanning the full temporal range of turbulence: the slow large-scale motions, the inertial range, and the fast dissipative scales.
- We define the sparse dictionary from a wavelet χ whose shape matches the oscillatory structure of the events, with 6 to 8 oscillations within its support (?). χ_m is defined from χ in order to cover the frequential domain with few overlap between the filters. This gives dilated wavelets

$$\chi_m(t) = \alpha_{j,q}\, \chi(2^{-m/3}t), \tag{17}$$

for $0 \leq m \leq M = 3J$. Because χ is more frequency-selective than the Morlet wavelet ψ, each covers only about a third of an octave, so three voices per octave are needed to tile the frequency axis without gaps.
- With this representation, for any extreme events, $|x{*}\chi_m(t)|$, bursts in a small region confined around (t) for a single j. (Fig. ??).
- The marginals of the coefficients χ_m are fat-tailed at high frequencies, where small-scale intermittency produces large-amplitude bursts; Gaussian at low frequencies, where large-scale fluctuations are near-normal; and Laplacian in between. The growth of the tails towards finer scales is a direct manifestation of increasing intermittency, large relative bursts occurring with non-negligible probability.
- Because of these events, the marginal distributions of Morlet wavelet coefficients are very sparse as well, and cannot be reproduced by only imposing $\mathbb{E}(\phi_1)$, $\mathbb{E}(\phi_2)$.
- We thus also include the wavelets ψ_j within the dictionary.

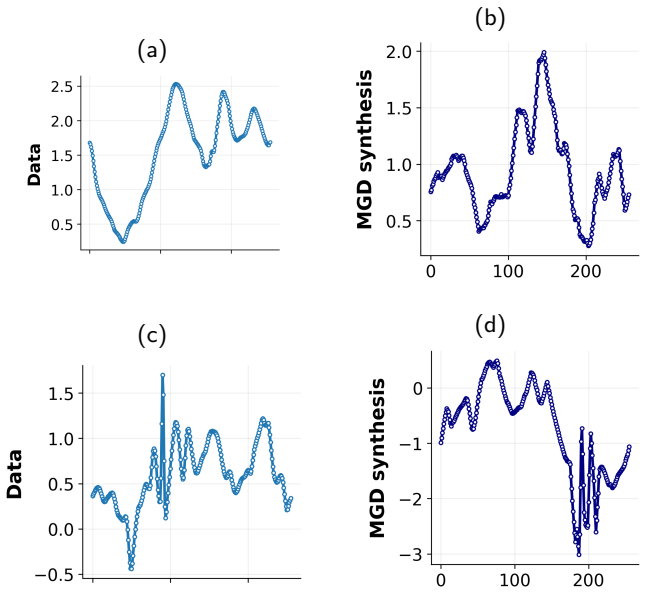


Figure 5 Regular Lagrangian turbulent path from the original dataset (c) and obtained with MGD (d). Lagrangian turbulent path exhibiting an extreme oscillatory event (highlighted in red) from the original dataset (e) and obtained with MGD (f).

Statistical validation of the model

SE REFERER AU PAPIER DE LUCA AINSI QUE CELUI DE PNAS NEXUS ET UTILISER LES MEME VALIDATIONS PAR MOMENTS. REGARDER LES EXPLICATIONS ET UTILISER POUR LA REDACTION.

- We adopt the statistical validations from (?)
- We conduct a validation on Figure ?? (Visual reproduction of rare events). In the case of lagrangian turbulence, it demonstrate that we can reproduce rare events of large amplitude.
- We conduct a statistical validation of the model, to verify that we are able to impose 1st, 2nd, 3rd and 4th order statistics from a reduced set of size $O(\log^3 d)$, instead of $O(d^3)$. We also show that our model extrapolate towards some higher order statistics, yet with some limits.
- We measure several statistics
  - Second order moments: Power Spectrum $\mathbb{E}(|\hat{x}(\omega)|^2)$
  - Histograms of increments $x(t+\tau) - x(t)$: strong intermittency is captured
  - Higher order structure function

$$S_p(\tau) = \mathbb{E}(|x(t+\tau) - x(t)|^p) \tag{18}$$

which measures how self-similar or not are scales in physics.

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– The local slope of the flatness, the golden standard of 4th order moments

$$\mathrm{Flat}_4(\tau) = \frac{\log \mathrm{S}_4(\tau)}{\log \mathrm{S}_2(\tau)}$$

It is a very precise measure of self-similarity. In Lagrangian turbulence, goes from a plateau at 2 at small τ (dissipative self similar regime) to a plateau of 1.6 at large τ (anomalous inertial self-similar regime), by dropping to lower value in between these two regimes, showing that turbulence is extremely not self similar.

– Time reversibility. From the discrete instantaneous power increment

$$\mathrm{dE}(\tau) = \big(x(t+\tau) - x(t)\big)x(t),$$

where $X(t)$ is the velocity field, we define

$$\mathrm{Sk}(\tau) = \frac{\mathbb{E}\big[(\mathrm{dE}(\tau) - \mathbb{E}[\mathrm{dE}(\tau)])^3\big]}{\mathbb{E}\big[(\mathrm{dE}(\tau) - \mathbb{E}[\mathrm{dE}(\tau)])^2\big]^{3/2}}, \tag{19}$$

which measures the skewness of the distribution of $\mathrm{dE}(\tau)$. Non-zero skewness indicates that the detailed micro balance is violated (detailed balance = the probability of going from energy E to energy $E + dE$ during a time period τ is equal to the probability to go from $E + dE$ to E.) and that the dynamic is not symetric.

– Higher order two-scale coefficient, which couples two lags τ and τ_*

$$C_{p,q}(\tau, \tau^\star) = \frac{\mathbb{E}\left[|x(t+\tau^\star) - x(t)|^p |x(t+\tau) - x(t)|^q\right]}{S_p(\tau) S_q(\tau^\star)}$$

where $C_{p,p} = C_{2p}$. Computed for $p = q = 2$, C_4. This coefficient is maximum for $\tau = \tau^*$. Physicists measure scales (loosely defined here) correlations by observing how fast C_4 decreases when τ varies away from τ^*. To capture interesting physics, τ^* should be chosen in the crossover turbulent range.

– High order statistics that are not prescribed, interactions beyond the 4th order or more than 2 points interactions $C_6(\tau, \tau^\star)$ and $C_{4,2}(\tau, \tau^\star)$

Table 1. Estimated negentropy (ΔH_*^σ) and normalising constants ($\log Z_\sigma$) computed for $\sigma = 3.5$, in the maximum entropy limit, for Eulerian and Lagragian turbulence.

Dataset	Negentropy	Normalizing Constant
Lagrangian Turbulence	3546.1 ± 2.2	-2575.7 ± 2.8

Table 2. Accuracy of the energy estimator, computed with Kullback-Leibler error $\mathbb{E}_{\hat{\theta}_n}\left[D_{\mathrm{KL}}(p_\theta \,\|\, p_{\hat{\theta}_n})\right]$ computed with $n = ?$ samples from MGD and Regularised MGD with the Monte Carlo sum (20). We compare with the theoretical Cramer Rao lower bound $r/2n$,

Dataset	MGD	Reg MGD	Cramer-Rao
Lagrangian Turbulence	1655.03	122.94	0.02

Accurate estimation of normalised energies in high dimension

- We verify numerically that our estimate of the energy is accurate, by estimating $\mathbb{E}\left[D_{\mathrm{KL}}(p^*_\theta \,\|\, p_{\hat{\theta}})\right]$ for estimators from MGD and regularised MGD.
- We compare Figure 2 Kullback-Leibler error $\mathbb{E}_{\hat{\theta}_n}\left[D_{\mathrm{KL}}(p_\theta \,\|\, p_{\hat{\theta}_n})\right]$ for estimators from MGD and Regularised, computed with the Monte Carlo sum

$$\mathbb{E}_{\hat{\theta}_n}\left[D_{\mathrm{KL}}(p_\theta \,\|\, p_{\hat{\theta}_n})\right] = \mathbb{E}_{\hat{\theta}}\left[\log \mathbb{E}_x\left[e^{-(\hat{\theta}_n - \theta)^\top \phi(x)}\right]\right]. \tag{20}$$

which can be estimated from data.
- We use the regularised energy to compute normalising constant values in table 1.
- Regularised MGD gains 2 order of magnitude in accuracy compared to MGD. Consistent with 1D multimodal experiments from previous section.
- We also compute the negentropy $S(g) - S(p)$, which measures non-Gaussianity of p as the entropy gap to the Gaussian g of same covariance as p. Using the MGD entropy bound s^*_σ, it is estimated by $\Delta S_\sigma = S(g) - S_\sigma$, where g has the same covariance as X.

Coupling parameters

Conclusion

This paper presents an algorithm to model, estimate and sample with accuracy from non gaussian energies. It gives access to a structured set of coupling coupling parameters between scales. The challenge is now to be able to link these coupling coupling parameters to the physics of the underlying processes, and connect it with phenomenological knowledge. In turbulence, it amounts to link the coupling parameters to cascade properties.

Appendices

- From equality to inequality proof for normalizing constant
- Explanation of scattering potentials
- Scattering Covariances
- Explanation of scalar potentials fitting wavelet coefficient marginals
- Accurate estimation of the Hamiltonian
- Sample-size reduction experiment

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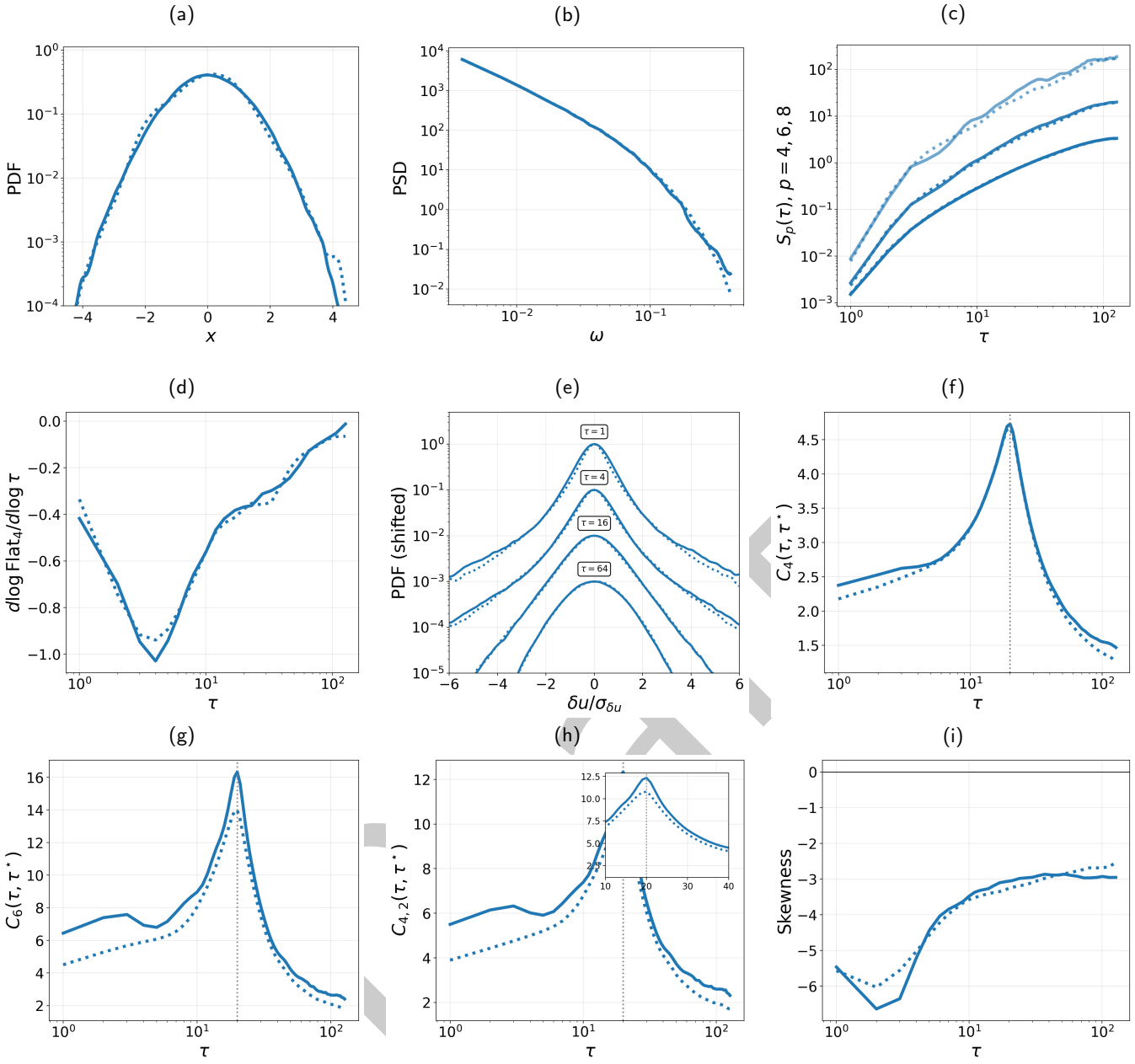


Figure 6 SPLIT INTO 3 FIGURES, LIKE IN LUCA, Each graph shows a specific family of moments estimated on the training data (dotted blue for Eulerian and dotted green for Lagrangian turbulence) and the estimated maximum entropy distribution, computed with MGD in the large $\sigma =?$ limit (solid blue for Eulerian and solid green for Lagrangian turbulence). (a) Marginal distributions of the signals. (b) Power spectrum. (c) Structure functions $\mathrm{S}_p(\tau)$ for $p = 4, 6, 8$. (d) Local slope of the flatness $\mathrm{Flat}_4(\tau)$. (e) PDFs of the increments $x(t + \tau) - x(t)$ at several scales τ, shifted by one decade per scale. (f) Fourth-order coefficient $C_4(\tau, \tau^\star)$. (g) Sixth-order coefficient $C_6(\tau, \tau^\star)$. (h) Mixed coefficient $C_{4,2}(\tau, \tau^\star)$ computed with $\tau^\star = 20$ (the inset magnifies the region around $\tau = \tau^\star$, marked by the dotted vertical line). (i) Skewness function $\mathrm{Sk}(\tau)$.

- Numerical details and algorithms (MGD DISCRETIZED ALGO AND ALL THE HYPERPARAMETERS USED+FIT OF THE MARGINALS IF NEEDED)

Scattering

Second order moments

- To reproduce spatial correlations, we want to include the covariance in the set of moments.
- Maximum entropy models conditioned on the covariance are gaussians.
- Assuming stationarity of the field, the covariance consists of moments

$$\mathbb{E}(x(t)x(t+\tau)) \text{ for } \tau = 0, ...d-1$$

The underlying moment function is of dimension $r = O(d)$, training thus requires at least $O(d)$ training samples, which is already prohibitive for high dimensional processes.

- The Fourier transform of the autocorrelation function $\mathbb{E}(x(t)x(t+\tau))$ is the power spectrum $\mathbb{E}|\hat{x}(\omega)|^2$. For multiscale processes, the power spectrum, which is also the eigenvalues of the covariance matrix, has a power law decay $\mathbb{E}|\hat{x}(\omega)|^2 \sim |\omega|^{-\eta}$. This can be approximated with logarithmic bining over diadic frequency bands $[2^{j-1}, 2^j]$. Dilated wavelets pave the frequency domain with few overlap in there support. Thus, the autocorrelation function is naturally approximated by the $O(\log d)$ moments $\mathbb{E}(\sum_t |\bar{x}_j(t)|^2)$, for $0 \leq j \leq J+1$.
- Said differently, the covariance matrix of multiscale stationary processes can be quasi diagonalised by wavelet basis or frame.
- Before we define the associated moment generating function, we must renormalise these coefficients. Multiscale random processes often have a power spectrum that has a power law decay, and so does the wavelet spectrum $\mathbb{E}(\sum_t |\bar{x}_j(t)|^2) \sim 2^{j\eta}$. If we consider a gaussian maximum entropy model build from moments $\mathbb{E}(\sum_t |\bar{x}_j(t)|^2)$, for $0 \leq j \leq J+1$, then the corresponding coupling parameters θ^* will scale like $2^{-j\eta}$, which varies over several order of magnitude for large signal, preventing any comparison, as θ^* coordinates corresponding to high frequencies will dominate. Furthermore, not normalising moment function leads to covariances matrices $\mathbb{E}(\phi\phi^\top)$ and $\mathbb{E}(\nabla\phi\nabla\phi^\top)$ that are ill-conditioned, leading to numerical instabilities. This may be avoided by normalising the moment function by

$$\sigma_j^2 = \mathbb{E}(\sum_t |\bar{x}_j(t)|^2)$$

, which rescales θ_* by σ_j^{-2}. The proposed normalization is closely related to Wilson's renormalization.
- Lead to a reduced set of $O(\log d)$ normalised moments computed from moment function

$$\phi_2(x) = \left(\sigma_j^{-2} d^{-1} \sum_t |\bar{x}_j(t)|^2\right)_{0 \leq j \leq J+1} \tag{21}$$

First order moments

- Intermittency, an important phenomenon in turbulence and finance, refers to the tendency of a random process to have activity that is not uniform but concentrated in localized, irregular bursts. It results from non-Gaussian distributions with fat tails and a peak at zero: most realizations are near zero, punctuated by a few large ones.
- In intermittent processes, the marginal distribution $p(\bar{x}_j(t))$ of wavelets coefficients is sparser than a gaussian distribution. This intermittency is standardly enforced at each scale by specifying a modulus moments $\mathbb{E}(|\bar{x}_j(t)|)$. Indeed the ratio of these with the second order moment $\mathbb{E}(|\bar{x}_j(t)|^2)$ gives a measure of sparsity.
- Lead to a reduced set of $O(\log d)$ normalised moments computed from moment function

$$\phi_1(x) = \left(\sigma_j^{-1} d^{-1} \sum_t |\bar{x}_j(t)|\right)_{0 \leq j \leq J+1} \tag{22}$$

- Strong intermittency might not be captured by ϕ_1 and ϕ_2, as maximum entropy distributions conditioned on $\mathbb{E}(\phi_1)$ and $\mathbb{E}(\phi_2)$ will have tails that decreases as fast as laplacian, and peak in 0 not sharper than Laplacian. This exclude rare events from this description, we shall come back to it later.
- The fourth order moment can have a long range dependencies. For stationary processes, these covariances matrices are diagonalised in the Fourier basis, and when the process is multiscale, the eigenvalues decrease with a power law. Thus, as for second order moments, these matrices are nearly diagonalized by applying a second wavelet transform.
- We may attempt a similar compression strategy for the third order moments. Here, because $Wx[j]$ has a limited support due to the convolution product with the wavelet ψ_j, we may only consider interactions $Wx[j]|Wx[j+k]| * \psi_j((t))$

Scattering Covariances

Scalar potentials

- We fit the full marginal histogram of $|Wx[j]|$ with a mixture of generalized Gaussian distributions, so that the potential adapts its exponent to the behaviour of the bulk and of the fat tails of the wavelet coefficients distribution, where the latter is the signature of the rare events of Lagrangian turbulence.
- For each scale j the amplitude axis $[0, \infty)$ is partitioned into $K_j \leq K$ regions by boundaries $0 < c_{j,1} < \cdots < c_{j,K_j-1} < \infty$. Boundaries are estimated from the data, initialised from quantiles and refined by a boundary search that minimises the composite negative log-likelihood over the whole histogram. Regions are not a hard partition: each is a soft window $w_{j,k}$ built from a sigmoid of $|Wx[j]| - c_{j,k}$ with a transition width proportional to the region width, so the windows sum to one along the amplitude axis.
- Within each region k we fit the truncated marginal density with a generalized Gaussian

$$p_{j,k}(t) \propto \exp\Big(-(|t|/s_{j,k})^{\alpha_{j,k}}\Big), \qquad t \in [c_{j,k-1}, c_{j,k}),$$

with shape $\alpha_{j,k}$ and scale $s_{j,k}$ estimated by maximum likelihood on the truncated segment. The normalising mass on $[c_{j,k-1}, c_{j,k})$ is computed from the incomplete gamma function, so the region boundaries enter the likelihood exactly rather than through a re-normalised global GGD.
- The lowest region captures the bulk of the histogram which contains the large majority of the probability mass, and is fit with a shape α close to the Gaussian or Laplacian range. The upper regions are fit independently, using only the rare, large-amplitude samples that fall inside them. This separates the tail fit from the bulk fit: a handful of extreme wavelet coefficients cannot bias the bulk shape, and the bulk cannot wash out the tail's own shape parameter, which is typically much smaller and signals a heavier tail.
- The corresponding potential is a windowed absolute moment

$$\phi_1^{(K)}(x) = \left(\sigma_j^{-1} \text{ Ave } w_{j,k}(|Wx[j]|)\, |Wx[j]|^{\alpha_{j,k}}\right)_{j,k},$$

one region-adapted moment per pair j, k. The exponent $\alpha_{j,k}$ is fit rather than fixed to one, so the potential matches the true tail decay of the histogram. This is what lets the

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potential sparsify rare events instead of only measuring their contribution to a linear average.
- We choose the number of active regions per scale adaptively from the empirical excess kurtosis of $Wx[j]$.
- We apply this construction to Morlet wavelets coefficients distributions and to the Q-band wavelet $\chi(t)$.

Accurate estimation of the hamiltonian

We derive the closed-form expressions for $\mathcal{L}_{\mathrm{DSM}}$ and $\mathcal{L}_{\mathrm{TSM}}$ used in Section 2. Neither derivation assumes that p_t is exponential; both rely only on integration by parts and on the identities

$$G_t\,\theta_t = \mathbb{E}\big[\Delta\phi(X_t)\big], \qquad G_t\,\eta_t = \frac{d}{dt}m_t, \tag{23}$$

where $G_t = \mathbb{E}\big[\nabla\phi(X_t)\cdot\nabla\phi(X_t)^\top\big]$ is the Gram matrix and θ_t, η_t are the MGD parameters.

Dual score-matching loss
Expanding the square in $\mathcal{L}_{\mathrm{DSM}}$ gives

$$\begin{aligned}\mathbb{E}\big[\|\tilde\theta_t^\top\nabla\phi + \nabla\log p_t\|^2\big] = \tilde\theta_t^\top G_t\tilde\theta_t \\ + 2\,\tilde\theta_t^\top\mathbb{E}\big[\nabla\phi(X_t)\cdot\nabla\log p_t(X_t)\big] \\ + \mathbb{E}\big[\|\nabla\log p_t(X_t)\|^2\big].\end{aligned}$$

The last term does not depend on $\tilde\theta$ and is absorbed into the constant. For the cross-term, integration by parts against p_t yields

$$\mathbb{E}\big[\nabla\phi(X_t)\cdot\nabla\log p_t(X_t)\big] = \int\nabla\phi\cdot\nabla p_t\,dx = -\mathbb{E}\big[\Delta\phi(X_t)\big],$$

so that

$$\mathcal{L}_{\mathrm{DSM}}(\tilde\theta) = \int_0^1\Big(\tilde\theta_t^\top G_t\tilde\theta_t - 2\,\tilde\theta_t^\top\mathbb{E}\big[\Delta\phi(X_t)\big]\Big)\,dt + \mathrm{cst}.$$

Time score-matching loss
Expanding the square in $\mathcal{L}_{\mathrm{TSM}}$ gives

$$\begin{aligned}\mathbb{E}\Big[\Big\|\tfrac{d}{dt}\tilde\theta_t^\top(\phi-m_t) + \tfrac{d}{dt}\log p_t\Big\|^2\Big] = \tfrac{d}{dt}\tilde\theta_t^\top C_t\tfrac{d}{dt}\tilde\theta_t \\ + 2\,\tfrac{d}{dt}\tilde\theta_t^\top\mathbb{E}\big[(\phi(X_t)-m_t)\tfrac{d}{dt}\log p_t(X_t)\big] \\ + \mathbb{E}\big[|\tfrac{d}{dt}\log p_t(X_t)|^2\big],\end{aligned}$$

where the last term is again absorbed into the constant. Using $p_t\frac{d}{dt}\log p_t = \partial_t p_t$ and the Fokker–Planck equation for X_t, an integration by parts in x transforms the cross-term into

$$\mathbb{E}\big[(\phi(X_t)-m_t)\,\tfrac{d}{dt}\log p_t(X_t)\big] = \frac{d}{dt}m_t,$$

since $\mathbb{E}[\phi(X_t)] = m_t$ along the MGD dynamics. This gives

$$\mathcal{L}_{\mathrm{TSM}}(\tilde\theta) = \int_0^1\Big(\tfrac{d}{dt}\tilde\theta_t^\top C_t\tfrac{d}{dt}\tilde\theta_t + 2\,\tfrac{d}{dt}\tilde\theta_t^\top\tfrac{d}{dt}m_t\Big)\,dt + \mathrm{cst},$$

with $C_t = \mathbb{E}\big[(\phi(X_t)-m_t)(\phi(X_t)-m_t)^\top\big]$.

Discretized MGD algorithm.
We follow the discretization of (**?**) (Sec. 3.2). The McKean–Vlasov SDE (**??**) is solved on a time grid $0 = t_0 < t_1 < \cdots < t_{n_t} = 1$, with steps $h_k = t_{k+1} - t_k$, by evolving an ensemble of n_{rep} interacting particles $(x_k^i)_{1\le i\le n_{\mathrm{rep}}}$ whose empirical measure approximates the law of X_{t_k}. All expectations over X_t appearing in (**??**)–(**??**) are replaced by empirical averages over the particles. The target moments $m_t = \mathbb{E}[\phi(I_t)]$ and their time derivatives are estimated from the n training samples $(x^j)_{1\le j\le n}$, each paired with a single Gaussian draw $z^j\sim\mathcal{N}(0,\mathrm{Id})$ that is fixed throughout the run. Writing $I_t^j = \cos(\alpha_t)\,z^j + \sin(\alpha_t)\,x^j$, we set

$$\hat m_t = \frac{1}{n}\sum_{j=1}^n\phi(I_t^j), \qquad \widehat{\dot m}_t = \frac{1}{n}\sum_{j=1}^n\nabla\phi(I_t^j)\cdot\dot I_t^j, \qquad \dot I_t^j = \dot\alpha_t\big(-\sin(\alpha_t)\,z^j - \cdots \tag{24}$$

so that $\widehat{\dot m}_t$ is the exact time derivative of $\hat m_t$. Each time step is split into a predictor and a corrector. This avoids evaluating $\mathbb{E}[\Delta\phi(X_t)]$, which is ill-defined for the non-smooth moment functions used here, such as moduli of wavelet coefficients.

Predictor. Given X_{t_k}, an Euler–Maruyama step of (**??**) with $\theta_t = 0$ gives

$$Y = X_{t_k} + h_k\,\eta_{t_k}^\top\nabla\phi(X_{t_k}) + \sqrt{2}\,\sigma\,(W_{t_{k+1}} - W_{t_k}), \qquad G_{t_k}\,\eta_{t_k} = \dot m_{t_k}. \tag{25}$$

This update does not preserve the moments: $\mathbb{E}[\phi(Y)] \neq m_{t_{k+1}}$.

Corrector. The moment condition is enforced by adding a term of the same form as the score term of (**??**),

$$X_{t_{k+1}} = Y - h_k\,\sigma^2\,\hat\theta^\top\nabla\phi(Y), \tag{26}$$

with $\hat\theta$ such that $\mathbb{E}[\phi(X_{t_{k+1}})] = m_{t_{k+1}}$. Solving this condition exactly is nonlinear in $\hat\theta$. It is also unnecessary, since the Euler–Maruyama scheme has weak order one in the step. Expanding to first order in h_k gives the linear system

$$h_k\,\sigma^2\,\mathbb{E}\Big[\nabla\phi(Y)\,\nabla\phi(Y)^\top\Big]\,\hat\theta = \mathbb{E}[\phi(Y)] - m_{t_{k+1}}. \tag{27}$$

To leading order in h_k, (27) reduces to $G_{t_k}\,\theta_{t_k} = \mathbb{E}[\Delta\phi(X_{t_k})]$. The corrector therefore implements the score term of the MGD SDE without computing $\Delta\phi$. Equation (27) also holds for empirical measures. As a result, the empirical mean of ϕ over the particles tracks $\hat m_{t_{k+1}}$ as the step size goes to zero, and a divergence of the particles would manifest as a moment mismatch (**?**).

Implementation choices. In our implementation, both linear systems are solved with a ridge regularization relative to the diagonal, $\hat G \to \hat G + \lambda\,\mathrm{diag}(\hat G)$. This is not part of the scheme of (**?**). With $\lambda > 0$, each corrector step removes the moment mismatch only approximately, and the residual enters the right-hand side of the next step. As discussed in (**?**), the drift scales as σ^2 for large σ, so the number of steps must grow as σ^2 and the computational cost of MGD scales as $O(\sigma^2)$. The time grid, σ, n_{rep} and λ used in our experiments are reported in Appendix **??**. The scheme is summarized in Algorithm 1.

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Algorithm 1 Moment Guided Diffusion (MGD), adapted from (?)

Require: volatility σ; time grid $(t_k)_{0\le k\le n_t}$ with $h_k = t_{k+1}-t_k$; number of particles n_{rep}; estimated moments $\hat{m}_{t_k}$ and time derivatives $\widehat{\dot{m}}_{t_k}$

Initialize $x_0^i \sim \mathcal{N}(0, \mathrm{Id})$ for $i = 1, \dots, n_{\mathrm{rep}}$

for $k = 0, \dots, n_t - 1$ **do**

Predictor

$\hat{G}_k = \frac{1}{n_{\mathrm{rep}}} \sum_{i=1}^{n_{\mathrm{rep}}} \nabla\phi(x_k^i)\, \nabla\phi(x_k^i)^\top$

Solve $\hat{G}_k\, \hat{\eta}_k = \widehat{\dot{m}}_{t_k}$ for $\hat{\eta}_k$

for $i = 1, \dots, n_{\mathrm{rep}}$ **do**

Sample $\xi_k^i \sim \mathcal{N}(0, \mathrm{Id})$

$y_k^i = x_k^i + h_k\, \hat{\eta}_k^\top \nabla\phi(x_k^i) + \sqrt{2h_k}\, \sigma\, \xi_k^i$

end for

Corrector (enforces the moment condition)

$\hat{G}'_k = \frac{1}{n_{\mathrm{rep}}} \sum_{i=1}^{n_{\mathrm{rep}}} \nabla\phi(y_k^i)\, \nabla\phi(y_k^i)^\top$

Solve $h_k\, \sigma^2\, \hat{G}'_k\, \hat{\theta}_k = \frac{1}{n_{\mathrm{rep}}} \sum_{i=1}^{n_{\mathrm{rep}}} \phi(y_k^i) - \hat{m}_{t_{k+1}}$ for $\hat{\theta}_k$

for $i = 1, \dots, n_{\mathrm{rep}}$ **do**

$x_{k+1}^i = y_k^i - h_k\, \sigma^2\, \hat{\theta}_k^\top \nabla\phi(y_k^i)$

end for

end for

Ensure: samples $(x_{n_t}^i)_{1\le i\le n_{\mathrm{rep}}}$

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Competing interest

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Data availability

Ethics statement

References

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